Autonomous Obstacle Avoiding Robot

Submitted By: Dharsshith Kumaran
Class: 5 D
PPS#: PPS2078

ABSTRACT

The **Autonomous Obstacle Avoiding Robot** is a two-wheel robotic vehicle designed to detect obstacles in its path and automatically select a suitable direction to continue its movement. The system is developed using an **Arduino microcontroller, ultrasonic sensor, servo motor, L293D motor driver, and two BO geared DC motors**. The robot demonstrates the practical application of embedded systems, sensor-based distance measurement, motor control, and real-time decision making.

An **HC-SR04 ultrasonic sensor** is mounted on a servo motor, allowing the sensor to scan different directions. During normal operation, the robot moves forward while continuously measuring the distance in front of it. The ultrasonic sensor is programmed with a maximum detection range of **200 cm**, and an obstacle is considered to be present when it is detected within **30 cm** of the robot.

When an obstacle is detected, the Arduino immediately stops both motors and commands the robot to move backward for **500 milliseconds**. The robot then stops and rotates the servo-mounted ultrasonic sensor toward the **left (160°)** and **right (20°)** directions. The distance measurements from both sides are obtained and compared. Invalid ultrasonic readings of zero are treated as the maximum detection distance to improve the decision-making process.

The robot uses a decision margin of **5 cm** when comparing the left and right distances. If the left side has significantly more free space, the robot turns left; if the right side has more free space, it turns right. When both directions provide approximately equal space, the robot follows a predefined default direction. If both sides are detected to be blocked within **20 cm**, the robot identifies the situation as a dead end and reverses for an additional **700 milliseconds** before attempting to continue.

The two BO motors are controlled through an **L293D dual H-bridge motor driver**. The Arduino controls the motor direction through four digital control pins and uses PWM outputs to drive the motors at **255 PWM**, providing maximum programmed motor speed. The servo motor is controlled through a dedicated Arduino pin, while the ultrasonic sensor uses separate trigger and echo pins. A serial monitor is also incorporated into the program to display real-time information such as front distance, obstacle detection, motor direction, left and right distance measurements, and the robot's navigation decision.

Components Used

- **Arduino board** – Main controller and decision-making unit.
- **HC-SR04 Ultrasonic Sensor** – Measures the distance to obstacles.
- **Servo Motor** – Rotates the ultrasonic sensor for left and right scanning.
- **L293D Motor Driver** – Controls the direction and speed of the two DC motors.
- **2 × BO Geared DC Motors** – Provide propulsion and differential steering.
- **Robot Chassis and Wheels** – Mechanical structure of the robot.
- **Battery/Power Supply** – Provides electrical power to the system.
- **Connecting Wires** – Used for electrical connections.
- **Serial Monitor** – Provides real-time system status and debugging information.

Applications

The basic technology demonstrated by this project can be used as a foundation for:

- Autonomous mobile robots
- Warehouse and material-handling robots
- Robotic vacuum cleaners
- Indoor delivery robots
- Security and surveillance robots
- Inspection robots
- Educational robotics platforms
- Search and exploration vehicles
- Automated navigation systems

Conclusion

The project demonstrates how an Arduino can integrate an ultrasonic sensor, servo motor, motor driver, and DC motors to create an autonomous navigation system. By scanning both sides of an obstacle and comparing the available distance, the robot is capable of making a basic real-time navigation decision rather than simply turning in a fixed direction. The project provides a practical foundation for developing more advanced autonomous robots incorporating additional sensors, wireless communication, mapping, and intelligent navigation algorithms.